AVALON ADVANCED MATERIALS

OTCQB: **AVLNF** | TSX: **AVL**

REDCHIP

Market Data

Avalon Advanced Materials Inc.
OTCQB: **AVLNF**
TSX: **AVL** | FRA: **OU5A**

Fiscal Year End	Aug. 31
Price[1]	$0.04
Market Cap[1]	$32.6M
Shares Out.[1]	834M
Float[1]	697M
Cash (mrq)[2]	$1.5M

Price & share data as of December 9, 2025

[1] stock price data in USD; OTCQB share data
[2] in Canadian Dollars

avalonadvancedmaterials.com

Company Overview

Avalon Advanced Materials is emerging as one of North America's most strategically positioned critical minerals companies, advancing a rare combination of high-value assets that directly support the continent's shift toward clean energy, advanced technologies, and secure supply chains. With a portfolio spanning rare earth elements, lithium, zirconium, tantalum, and niobium, Avalon offers investors exposure to some of the fastest-growing and most geopolitically important critical minerals in the world amid a secular shift toward reshoring the production of such feedstocks.

Anchored by the Nechalacho Rare Earth Elements and Zirconium Project—one of the largest and highest quality North American resources containing both light and heavy rare earths—Avalon is positioned to become a foundational supplier to industries ranging from defense and communications to EV motors and renewable energy systems. Complementing this rare earth strength, the Company has plans to build Ontario's first midstream lithium processing hub via a state-of-the-art lithium hydroxide facility in Thunder Bay designed to supply the booming EV battery manufacturing and energy storage system ecosystem across Canada and the U.S. .

Through its JV with global industrial minerals leader SCR-Sibelco, Avalon is also rapidly expanding its lithium resource base at Separation Rapids —reinforcing a highly scalable, multi-asset growth platform. With renewed leadership, tier-one partnerships with companies such as Metso and Qualcomm, and recent financing that accelerates development across its portfolio, Avalon is transitioning into a pivotal supplier of critical materials at a time when North America is aggressively reshoring and securing its mineral supply chains. With a market cap a fraction of the estimated net present value of its rare earth and lithium assets, Avalon is uniquely positioned to unlock substantial long-term value as management executes and demand for rare earths and lithium continues to surge.

Investment Thesis

Multiple Value Creation Levers

- Management reset
- Board refresh: ~60% new directors
- Strengthening capital structure
- Further resource validation
- Securing strategic funding
- Additional strategic partnerships
- Non-core asset monetization
- Broadening of shareholder base

Unique Diverse Portfolio

- Nechalacho REE DFS (2013) - After-Tax NPV @ 8%: $1.3B and 19.6% IRR
- Thunder Bay Facility PEA (2024) After-Tax $4.1B NPV & 48% IRR
- Separation Rapids - JV Sibelco

High Demand Markets

Strategically positioned in rapidly growing markets such as electric vehicles, renewable energy storage, and advanced technologies.

Strong Partnerships

Collaborations with leading industry players to enhance our capabilities and market reach, fostering innovation and growth:

- JV with Sibelco
- Partnership with Metso Corp.
- Collaboration with Qualcomm

Government Alignment

Advancing rare earth and lithium assets central to national defense, technological advancement, energy and industrial policy (strengthening critical mineral supply chains).

Valuation Disconnect

- Market cap materially below intrinsic asset values (REE $1.2B NPV, Lithium $4.1B NPV)
- Management committed to closing this gap through disciplined execution against multiple value creation levers

Investment Highlights

Premier North American Critical Minerals Platform with Multi-Asset Upside

- Portfolio anchored by Nechalacho, one of the largest, highest quality North American rare earth deposits containing both light and heavy REEs, plus zirconium, tantalum, and niobium.
- First mover in Ontario's midstream lithium processing, bridging upstream resources with North America's booming EV battery market.

Exceptional Project Economics Supporting Significant Value Re-Rating Potential

- Lake Superior Lithium's PEA (2024) projects C$4.1B after-tax NPV (8%), 48% IRR, and 2.5-year payback, with upcoming Feasibility Study aimed at further refining the engineering and economics.
- Nechalacho DFS (2013) outlines C$1.26B after-tax NPV and 19.6% IRR, with a validated flowsheet and substantial strategic importance for defense and clean energy applications.

Backed by Tier-One Partners and Fully Aligned with Government Policy Priorities

- Strategic partnerships with Metso and Qualcomm Technologies enhance technological leadership and operational readiness.
- Sibelco JV brings global industrial minerals expertise and committed exploration capital.
- Avalon's projects directly support national goals in clean energy, electrification, supply chain security, and defense readiness.

Multiple High-Impact Catalysts to Drive Shareholder Value

- Nechalacho DFS refresh commissioned to revalidate economics and unlock strategic funding and partnership opportunities.
- Lake Superior Lithium Feasibility Study commissioned to validate economics, with feedstock and offtake agreements as targeted near-term milestones.
- Exploration and resource expansion across the Separation Rapids region via Sibelco-funded programs.
- Monetization of non-core assets (tin, anorthosite) offers prospect of non-dilutive capital for growth.

Among First Midstream Lithium Processors in Ontario: A Strategic Gateway to North American EV Growth

- Thunder Bay facility expected to be Ontario's first commercial lithium hydroxide refineries, providing domestic supply for the rapidly expanding Canadian and U.S. battery ecosystem.
- Well-positioned with direct access to rail, deep-water port, and major highways, enabling efficient logistics from feedstock to end users.
- Partnership with Metso supports a cutting-edge, acid-free alkaline leach process for high-purity battery-grade lithium hydroxide production.

Exposure to High-Demand Markets Facing Increasing Supply Constraints

- Lithium demand expected to grow sharply through 2035, with Western refining capacity heavily undersupplied.
- Rare earths remain mission-critical for EV motors, wind turbines, smartphones, aerospace, and defense systems.
- Strong structural tailwinds from Canadian and U.S. incentives supporting domestic critical mineral development.

Strengthened Leadership and Governance Accelerating Execution

- Corporate reset with new senior leadership and ~60% refreshed board, adding deep mining, finance, and capital markets expertise.
- Focused on disciplined execution: resource expansion, project de-risking, strategic partnerships, and capital efficiency.
- Strengthened balance sheet following recent financing to accelerate development across core assets.

Value Proposition

Avalon Advanced Materials offers investors a unique opportunity to participate in the emergence of a multi-asset, strategically vital North American critical minerals platform at a moment when governments and industry are racing to regionalize supply chains for rare earths and lithium. With one of the only deposits in North America hosting both light and heavy rare earth elements alongside zirconium, tantalum, and niobium, Avalon is positioned to become a foundational supplier to sectors spanning electric vehicles, wind energy, defense, and advanced communications. At the same time, the Company has plans to build Ontario's first commercial lithium hydroxide refinery—a midstream asset capable of supplying the rapidly expanding EV battery manufacturing hubs across Canada and the United States. This dual-asset strategy provides diversification and creates multiple avenues for value creation: upstream resource expansion, midstream processing, strategic partnerships, and exposure to structural shortages in both lithium and rare earths.

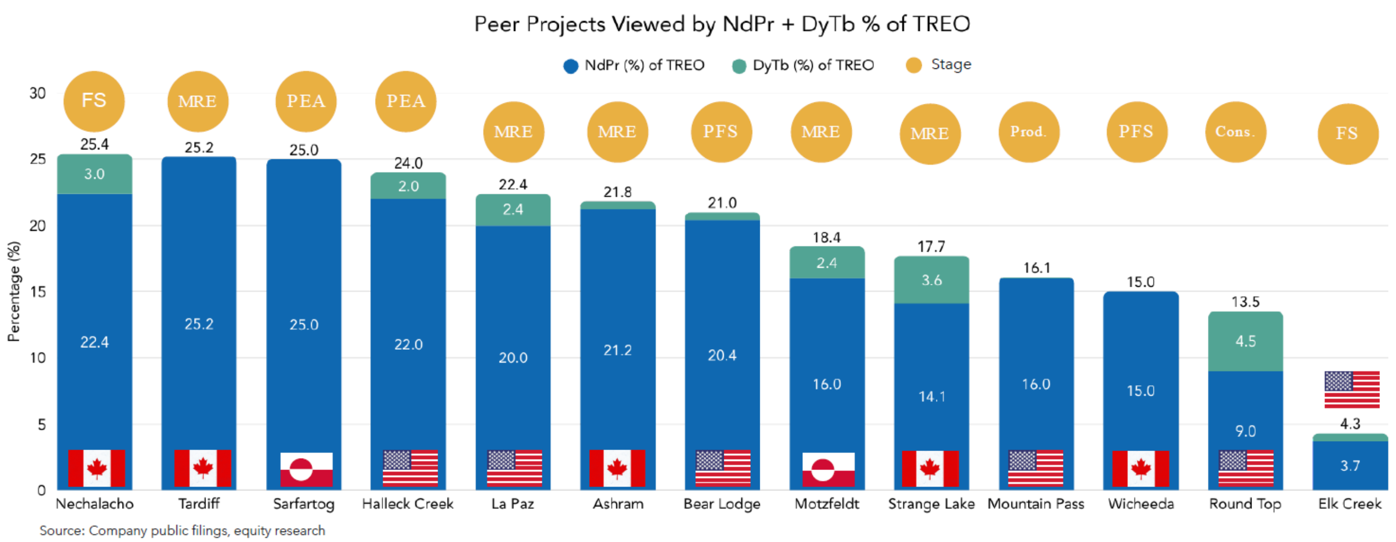
Avalon's recent corporate reset—including a strengthened leadership team, refreshed board, and reinforced capital structure—provides the operational discipline and execution capability needed to unlock the full potential of its asset portfolio. Strategic partnerships with SCR-Sibelco NV, Metso, Qualcomm Technologies, and Engina Corporation further amplify Avalon's competitive positioning by bringing world-class processing expertise, exploration funding, and technology collaboration to its core projects. With major catalysts ahead—from the Nechalacho DFS refresh and Lake Superior Lithium's progression to Feasibility Study, the prospect of additional strategic partnerships —the Company is poised to begin narrowing the substantial disconnect between its current market capitalization and the intrinsic value indicated by independent economic studies, including Nechalacho's DFS and the Thunder Bay refinery's C$4.1 billion after-tax NPV.

In a supply environment increasingly defined by geopolitical tension, government intervention, and accelerating demand for clean energy technologies, Avalon stands out as a scarce, strategically aligned platform with the potential to become a long-term domestic supplier of materials essential to electrification and national security. For investors, Avalon represents a differentiated opportunity with asymmetric upside: exposure to high-growth critical mineral markets, backed by premium assets, strategic partnerships, and strong policy tailwinds—at an inflection point where disciplined execution can translate into significant shareholder value.

Critical Rare Earths Applications in Technology

Highest Grade & Among Most Advanced Development Paths in Rare Earth Peer Group

One of few projects to have achieved DFS-level – with unmatched quality (NdPr /DyTb intensity)



Asset Locations

Office Headquarters
Toronto, ON

Lake Superior Lithium Inc.
Thunder Bay, ON

Separation Rapids Ltd.
Kenora, ON

Nechalacho Project
Thor Lake, NWT

East Kemptville
Yarmouth, NS

Warren Township Project
Timmins, ON